

Vaibhav Deshmukh

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Academic Qualifications

- **Master of Science - Mechanical Engineering**
Arizona State University (3.82/4) **Aug 2015- May 2017**
- **Bachelor of Technology - Mechanical Engineering**
VNIT India (8.38/10) **Aug 2010- May 2014**

Research Publications

- Research publication on design and simulation of control algorithms for particle sized ground robots. **Dec 2017**
 - Link to Research [Paper1](#)
- Research publication on practical sub-optimal implementation of the control algorithm on palm sized ground robots. **Jan 2018**
 - Link to Research [Paper2](#)

Strengths

- Matlab, Simulink, Python, Cpp
- Controller Design, System Identification, State Estimation Algorithms
- Control Theory, Kalman Filter, System Identification
- Machine Learning Frameworks- Scikit, TensorFlow
- Experience with Embedded systems, Auto coding, Agile development.
- Experience in MIL (Model in Loop) and SIL (Software in Loop) testing.
- Git, Plastic, Wiki, Jira, Crucible.
- Hardware Test Tools- CANalyzer, Canoe
- ROS, Turtlebot, Gazebo, Carmaker, Dspace Micro Autobox

Professional Work Experience

- 1. Automated Driving Algorithm Developer** **Nov 2021- Current Date**
Aptiv Corporation (4.5 years)

Role 3- Motion Controls- Algorithm Engineer (3 years)

April 2023- Current Date

- Lead development of Lateral Velocity based Lane Keep Assist algorithm for production vehicles
- Tuned Model Predictive Controller and PID controllers in vehicle for production grade software- handling real time and performance constraints (closed-loop and open-loop controls)
- Lead development for converting front wheel steer control algorithms to be compatible with 4-wheel steer systems
- Develop and maintain Level 2 and Level 2 Plus Highway Assist and Lane Keep Assist Features

- Blackbox System Identification and Deployment- Converted customer provided black box model into transfer function models and deployed in production vehicles
- Dspace, Embedded and Carmaker based vehicle simulations testing of ADAS Algorithms and Features
- Developed and validated Trailer Stability and Oscillation Estimation for Highway Assist with Trailer Feature
- Developed universal resimulation scripts capable of handling multiple customer software

Role 2- Vehicle State Estimation- Algorithm Engineer (1 year)

April 2022- July 2023

- Led the development and testing of trailer mass estimation software
- Debugging and resolving vehicle state estimation errors in production software
- Developed regression framework for validating new algorithm development
- Manage software release process and fix integration issues (C++, C)

Role 1- Mode Management- Algorithm Engineer (0.5 year)

Nov 2021 - April 2022

- Designed Simulink based algorithms and State machines that handle safety critical Level 2 mode manager
- Implement algorithms that manage system degradation and ODD related system failures
- Create Simulink based Test cases to enable continuous system validation and verification
- Validate and assess algorithm performance on real and synthetic data
- Support on-road testing and data collection efforts

2. Control Systems Engineer

Apr 2018- Oct 2021

Caterpillar Trimble Control Technologies (3.5 years)

- System Identification and Analysis for Hydraulic Cylinder First order system
- Designed algorithm for Machine Forward Kinematics and Angle Calculation based on Euler rotations.
- Resolving run-time errors for Control Models- Math errors, fault in Rest Detection logic, Blade Angle errors, etc.
- Writing Python, C++ unit tests for measure-up and vector math for machines
- Debugging state estimator and controller errors observed during field testing using Matlab, Simulink.
- Designed MATLAB apps/tools to aid in data collection and processing.

3. Robotics Engineering Intern

Nov 2017 – Apr 2018

Mitsubishi Electric Research Lab, Cambridge (0.5 year)

- Researched the prevalent algorithms for implementation on real world robots
- Implemented Python based scripts for implementation on Gazebo and Turtlebot robotic environment
- Validation and Implementation of algorithm on three Turtlebots equipped with Lasers, Kinect and Cameras- in Linux Environment

4. Graduate Research Engineer- Control Systems

Nov 2017 – Oct 2018

ASU Autonomous Collective Systems Laboratory Lab, Tempe (1 year)

- Researched the prevalent algorithms and techniques used in small-sized robots.
- Simulated the control algorithms in simulink and gazebo robotic simulation platform

- Researched and published two research papers on developing novel control algorithm in the field of swarm robotics.
- Implemented these control algorithms on small sized ground robots.

5. Graduate Engineer Trainee

July 2014 – June 2015

Hero MotoCorp Ltd., India (1 year)

- Analyzed vehicle logs, created performance models and used data fitting techniques for analysis and coverage reports.
- Conducted performance testing on bikes, including brake test, speed test, acceleration tests.